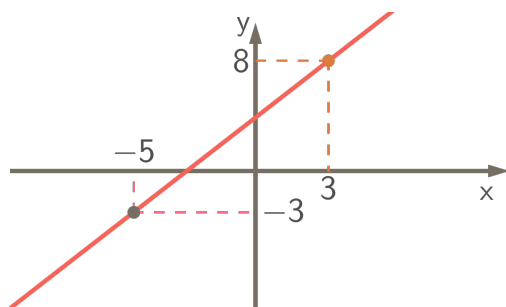


01

Encontre o coeficiente angular da reta:



- (A) $\frac{22}{5}$
- (B) $\frac{11}{8}$
- (C) $\frac{5}{2}$
- (D) $\frac{3}{10}$

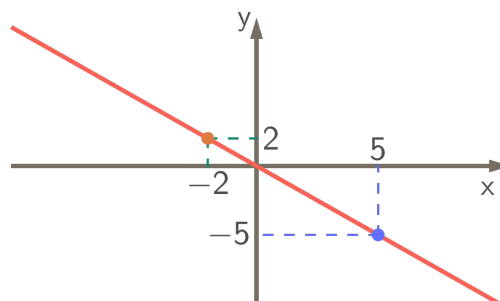
03

Determine os valores de x para os quais $f(x) = 4x + 32 < 0$.

- (A) $x > -8$
- (B) $x < -8$
- (C) $x \geq -8$
- (D) $x \leq -8$

02

Encontre o valor de b : $f(x) = \frac{-7}{7}x + b$



- (A) -1
- (B) 0
- (C) -2
- (D) -3

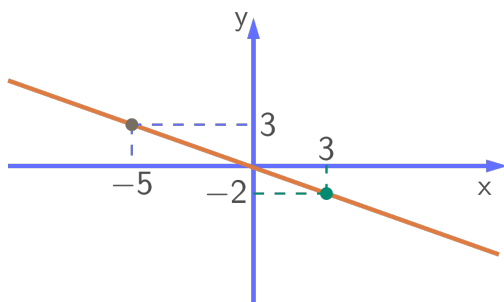
04

Determine os valores de x para os quais $f(x) = 5x + 10 < 0$.

- (A) $x \leq -2$
- (B) $x < -2$
- (C) $x > -2$
- (D) $x \geq -2$

05

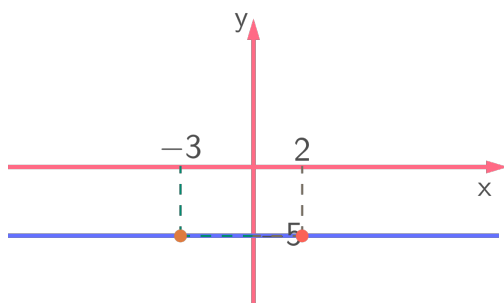
Encontre o coeficiente angular da reta:



- (A) $-\frac{2}{5}$
- (B) $-\frac{5}{8}$
- (C) $-\frac{21}{26}$
- (D) $-\frac{11}{2}$

07

Encontre o valor de b : $f(x) = 0x + b$



- (A) -6
- (B) -4
- (C) -8
- (D) -5

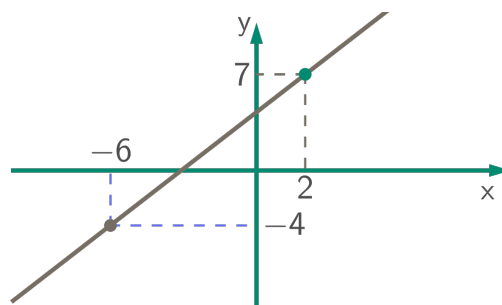
06

Determine os valores de x para os quais $f(x) = 4x + 12 > 0$.

- (A) $x \leq -3$
- (B) $x < -3$
- (C) $x > -3$
- (D) $x \geq -3$

08

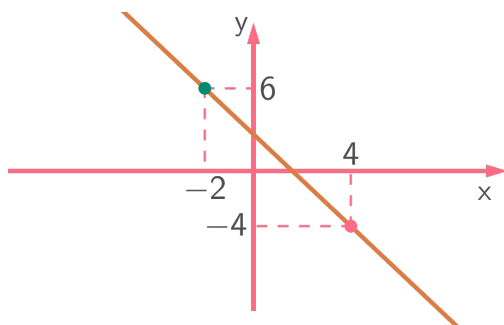
Encontre o valor de b : $f(x) = \frac{11}{8}x + b$



- (A) $\frac{17}{4}$
- (B) $\frac{16}{7}$
- (C) $\frac{1}{2}$
- (D) $\frac{8}{15}$

09

Encontre o coeficiente angular da reta:



- (A) 7
- (B) $\frac{24}{17}$
- (C) 3
- (D) $-\frac{5}{3}$

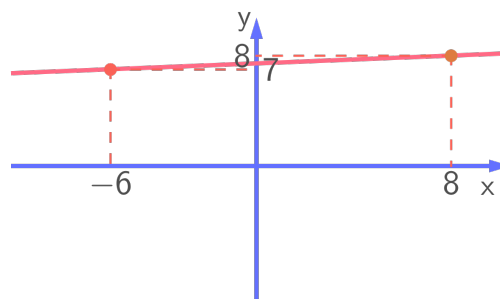
11

A função afim $f(x) = ax + b$ passa pelos pontos $A = (-5, -1)$ e $B = (5, 1)$. Calcule $f(2)$.

- (A) $\frac{13}{14}$
- (B) 5
- (C) $\frac{17}{3}$
- (D) $\frac{2}{5}$

10

Encontre o valor de b : $f(x) = \frac{1}{14}x + b$



- (A) $\frac{52}{7}$
- (B) 2
- (C) $\frac{43}{2}$
- (D) $\frac{46}{27}$

12

A função afim $f(x) = ax + b$ passa pelos pontos $A = (-2, 6)$ e $B = (2, -5)$. Calcule $f(3)$.

- (A) $-\frac{50}{3}$
- (B) $-\frac{13}{6}$
- (C) $-\frac{37}{7}$
- (D) $-\frac{31}{4}$

13

Determine os valores de x para os quais $f(x) = 5x - 15 > 0$.

(A) $x > 3$

(B) $x \leq 3$

(C) $x \geq 3$

(D) $x < 3$

15

A função afim $f(x) = ax + b$ passa pelos pontos $A = (-3, -3)$ e $B = (3, 6)$. Calcule $f(4)$.

(A) $\frac{5}{9}$

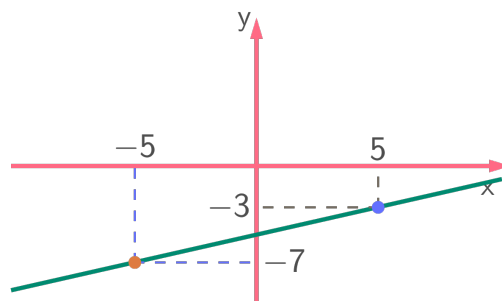
(B) $\frac{15}{2}$

(C) $\frac{1}{9}$

(D) $\frac{7}{4}$

14

Encontre o valor de b : $f(x) = \frac{4}{10}x + b$



(A) -5

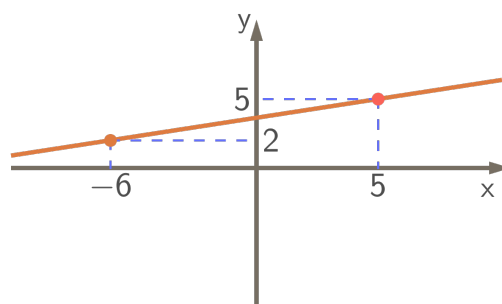
(B) -3

(C) -4

(D) -7

16

Encontre o valor de b : $f(x) = \frac{3}{11}x + b$



(A) $\frac{7}{3}$

(B) $\frac{21}{31}$

(C) $\frac{29}{8}$

(D) $\frac{40}{11}$

01 . B

Dados os pontos $A = (-5, -3)$ e $B = (3, 8)$, temos que o coeficiente angular a é calculado por:

$$a = \frac{\Delta y}{\Delta x} = \frac{y_B - y_A}{x_B - x_A}$$
$$a = \frac{8 - (-3)}{3 - (-5)} = \frac{8 + 3}{3 + 5} = \frac{11}{8}$$

03 . B

Primeiro encontramos o zero x_0 :

$$4x + 32 = 0$$

$$x_0 = -\frac{32}{4} = -8$$

A função é crescente ($a > 0$), portanto:

$$f(x) < 0 \text{ quando } x < -8$$

05 . B

Dados os pontos $A = (-5, 3)$ e $B = (3, -2)$, temos que o coeficiente angular a é calculado por:

$$a = \frac{\Delta y}{\Delta x} = \frac{y_B - y_A}{x_B - x_A}$$
$$a = \frac{-2 - 3}{3 - (-5)} = \frac{-2 - 3}{3 + 5} = \frac{-5}{8}$$

07 . D

O coeficiente linear b pode ser calculado por:

$$y = 0x + b$$

Considerando o ponto $(-3, -5)$, temos:

$$-5 = 0 \cdot -3 + b$$

$$-5 = 0 + b$$

$$-b = 0 + 5 = 5$$

$$b = -5$$

02 . B

O coeficiente linear b pode ser calculado por:

$$y = -\frac{7}{7}x + b$$

Considerando o ponto $(-2, 2)$, temos:

$$2 = -\frac{7}{7} \cdot -2 + b$$

$$2 = 2 + b$$

$$-b = 2 - 2 = 0$$

$$b = 0$$

04 . B

Primeiro encontramos o zero x_0 :

$$5x + 10 = 0$$

$$x_0 = -\frac{10}{5} = -2$$

A função é crescente ($a > 0$), portanto:

$$f(x) < 0 \text{ quando } x < -2$$

06 . C

Primeiro encontramos o zero x_0 :

$$4x + 12 = 0$$

$$x_0 = -\frac{12}{4} = -3$$

A função é crescente ($a > 0$), portanto:

$$f(x) > 0 \text{ quando } x > -3$$

08 . A

O coeficiente linear b pode ser calculado por:

$$y = \frac{11}{8}x + b$$

Considerando o ponto $(-6, -4)$, temos:

$$-4 = \frac{11}{8} \cdot -6 + b$$

$$-4 = -\frac{33}{4} + b$$

$$-b = -\frac{33}{4} + 4 = -\frac{17}{4}$$

$$b = \frac{17}{4}$$

09 . D

Dados os pontos $A = (-2, 6)$ e $B = (4, -4)$, temos que o coeficiente angular a é calculado por:

$$a = \frac{\Delta y}{\Delta x} = \frac{y_B - y_A}{x_B - x_A}$$

$$a = \frac{-4 - 6}{4 - (-2)} = \frac{-4 - 6}{4 + 2} = \frac{-10}{6} = \frac{-5}{3}$$

11 . D

Coeficiente angular a :

$$a = \frac{\Delta y}{\Delta x} = \frac{y_B - y_A}{x_B - x_A}$$

$$a = \frac{1 - (-1)}{5 - (-5)} = \frac{1 + 1}{5 + 5} = \frac{2}{10} = \frac{1}{5}$$

Coeficiente linear b usando $A = (-5, -1)$:

$$-1 = \frac{1}{5} \cdot (-5) + b$$

$$b = -1 + 1 = \mathbf{0}$$

Logo $f(x) = \frac{1}{5}x$:

$$f(2) = \frac{1}{5} \cdot 2 + 0 = \frac{2}{5} = \frac{\mathbf{2}}{\mathbf{5}}$$

13 . A

Primeiro encontramos o zero x_0 :

$$5x - 15 = 0$$

$$x_0 = \frac{15}{5} = \mathbf{3}$$

A função é crescente ($a > 0$), portanto:

$$f(x) > 0 \text{ quando } \mathbf{x > 3}$$

10 . A

O coeficiente linear b pode ser calculado por:

$$y = \frac{1}{14}x + b$$

Considerando o ponto $(-6, 7)$, temos:

$$7 = \frac{1}{14} \cdot -6 + b$$

$$7 = -\frac{3}{7} + b$$

$$-b = -\frac{3}{7} - 7 = -\frac{52}{7}$$

$$b = \frac{52}{7}$$

12 . D

Coeficiente angular a :

$$a = \frac{\Delta y}{\Delta x} = \frac{y_B - y_A}{x_B - x_A}$$

$$a = \frac{-5 - 6}{2 - (-2)} = \frac{-5 - 6}{2 + 2} = \frac{-11}{4}$$

Coeficiente linear b usando $A = (-2, 6)$:

$$6 = \frac{-11}{4} \cdot (-2) + b$$

$$b = 6 - \frac{11}{2} = \frac{\mathbf{1}}{\mathbf{2}}$$

Logo $f(x) = \frac{-11}{4}x + \frac{1}{2}$:

$$f(3) = \frac{-11}{4} \cdot 3 + \frac{1}{2} = \frac{-33}{4} + \frac{1}{2} = \frac{-\mathbf{31}}{\mathbf{4}}$$

14 . A

O coeficiente linear b pode ser calculado por:

$$y = \frac{4}{10}x + b$$

Considerando o ponto $(-5, -7)$, temos:

$$-7 = \frac{4}{10} \cdot -5 + b$$

$$-7 = -2 + b$$

$$-b = -2 + 7 = 5$$

$$b = -5$$

15 . B

Coeficiente angular a :

$$a = \frac{\Delta y}{\Delta x} = \frac{y_B - y_A}{x_B - x_A}$$

$$a = \frac{6 - (-3)}{3 - (-3)} = \frac{6 + 3}{3 + 3} = \frac{9}{6} = \frac{3}{2}$$

Coeficiente linear b usando $A = (-3, -3)$:

$$-3 = \frac{3}{2} \cdot (-3) + b$$

$$b = -3 + \frac{9}{2} = \frac{3}{2}$$

Logo $f(x) = \frac{3}{2}x + \frac{3}{2}$:

$$f(4) = \frac{3}{2} \cdot 4 + \frac{3}{2} = 6 + \frac{3}{2} = \frac{15}{2}$$

16 . D

O coeficiente linear b pode ser calculado por:

$$y = \frac{3}{11}x + b$$

Considerando o ponto $(-6, 2)$, temos:

$$2 = \frac{3}{11} \cdot -6 + b$$

$$2 = -\frac{18}{11} + b$$

$$-b = -\frac{18}{11} - 2 = -\frac{40}{11}$$

$$b = \frac{40}{11}$$